Measurement Lab Report

Name: Adrien Jefferson **Partner:** _____ **Section #:** ___

Data Sheet (should be copied into Lab Notebook)

Identifying an Unknown Metal

1. Volume Measurements for Various Liquids (1 point each box)

In Table 2, record the volume of liquid in the containers as directed by your Lab Instructor. Use the correct number of decimal places and units. Check with your instructor before proceeding to the next section.

Table 2. Volumes

	Volume of Liquid
A	19.1 mL
B	30.7 mL
C	70.1 mL
D	40.8 mL

2. Density of Unknown Metal Determined with Volume by Calculation (2 points each blank)

a. Unknown metal ID # 4

b. Mass of unknown metal = 19.937 g

c. Dimension #1 of unknown metal = Radius = 0.60 cm

d. Dimension #2 of unknown metal = Height = 1.80 cm

e. Dimension #3 of unknown metal = ———

f. Calculate the volume of the unknown metal. (Show calculation) (3 points)

$V = \pi r^2 h$

$V = \pi\ 0.60^2_{cm}(1.80)cm$

$V = 0.904$

Volume of Metal by Calculation: 0.90 cm

g. Calculate the density of the unknown metal. (Show calculation) (3 points)

$D = \frac{m}{V}$ $D = \frac{19.937g}{0.90\ cm}$ $D = 22.152$

Density of Metal using Volume by Calculation: 22.1 g/mL

3. Density of Unknown Metal Determined with Volume by Displacement Method (2 points each)

a. Initial Level of Water in Graduated Cylinder: 20 mL

b. Final Level of Water in Graduated Cylinder after adding Metal: 22 mL

c. Calculate the volume of the unknown metal determined with volume by the displacement method. (Show calculation) (3 points)

22 - 20

Volume of Metal using the Displacement Method: 2.0 cm

d. Calculate the density of the unknown metal using the volume by displacement method. (Show calculation) (3 points)

$\frac{19.937}{2.0}$

Density of Metal using the Volume by Displacement Method: 9.0 mL

Appearance of the Metal

e. Describe the appearance of the unknown metal. Use qualitative descriptors such as size, shape, luster (how shiny it is), hardness, etc. (2 points)

Unknown metal is shaped like a cylinder, shines as if its made of silver alloy and is as hard as a metal screw.

4. Measurements an Unknown Solution using a Buret

a. Unknown Solution ID # 1 (1 point)

b. Record data below, using the correct number of decimal places and units.
(1 point each blank)

Table 3. Measurements of an Unknown Solution using a Buret

	Trial #1	Trial #2	Trial #3*
Mass of clean empty beaker	31.515 g	31.515 g	31.515 g
Mass of beaker + solution	41.282 g	44.290 g	46.616 g
Mass of solution	9.767 g	12.775 g	15.101 g
Initial Level in Buret	29.9 mL	19.1 mL	14.5 mL
Final Level in Buret	38.8 mL	31.2 mL	29.9 mL
Volume of Solution delivered	8.90 mL	12.1 mL	14.4 mL
Density of Solution	1.10 mL	1.06 mL	1.05 mL
Average Density of Solution	(1.10 mL + 1.06mL)/2 = 1.08 mL		
Percent Difference	3.70 mL		

*Carry out a third trial but calculate the percent difference between Trial #1 and Trial #2 only.

c. Calculate the percent difference in density for Trial #1 and Trial #2 using the formula below. (Show your calculation) Note: Because percent difference is the ***absolute value*** of the difference in the two density values divided by the average, it is always ***positive***. (3 points)

$$\text{percent difference} = \frac{|Trial\#1 - Trial\#2|}{Average\ of\ Trials\ \#1\ and\ \#2} \times 100$$

Percent Difference = $\frac{(1.10\text{mL} - 1.06\text{mL})}{1.08\text{ mL}} \times 100 = 3.70\text{ mL}$

Mass of Solution

41.282 − 31.515

44.290 − 31.515

46.616 − 31.515

Lab Report (Post-lab):

1. Identifying an Unknown Metal (2 points each)

a. Transfer the densities you calculated on page 9 to the appropriate places below:

Density of the Unknown Metal Using Volume by Calculation (#2g above): 0.90 cm	Density of the Unknown Metal Using Volume by Displacement Method (#3d above): 2.0 cm

b. Compare the density of your unknown metal to those in Table 4 below:

Table 4. Densities of Common Substances

Substance	Density (g/cm^3)	Substance	Density (g/cm^3)
Gold	19.300	Iron	7.860
Lead	11.400	Stainless Steel	7.740
Silver	10.500	Titanium	4.510
Zinc	7.140	Aluminum	2.700
Brass	8.690	Ice (0°C)	0.917
Copper	8.960	Water (at 4°C)	1.000

c. Identify your metal by comparing your calculated density to the densities provided in Table 4 above:

Displacement = Copper Calculated = Gold ?

d. Explain how the *appearance* of the metal fits the identity of the metal, as determined by density.

As for Gold it doesn't fit appearence when it comes to color and hardness. for Displacement if the metal is mixed with Copper and another metal then the grey/silver color makes sense.

e. Show your calculation for the Percent Error for each density below: (Note: Because percent error is the *absolute value* of the difference of the two values divided by the true value, percent error is always positive.) (4 points total)

$$\text{Percent Error} = \left| \frac{\text{Experimental Density of Metal} - \text{True Density for Metal from Table}}{\text{True Density for Metal from Table}} \right| \times 100$$

~~9.0~~ 22 g/cm

Percent Error = $\frac{22\ g/cm - 19.3\ g/cm}{19.3\ g/cm} \times 100 = 14\ g/c$

$\frac{9.0\ g/cm - 8.960\ g/cm}{8.960\ g/cm} \times 100 = 0.45$

Percent Error for Density using Volume by Calculation: 14

Percent Error for Density using Volume by Displacement Method: 0.45

f. Which method (density using volume by calculation or density using volume by displacement method) resulted in a higher percent error and a less accurate density? Is this what you would expect? Explain possible sources of experimental error. (Complete sentences) (3 points)

Density by calculation resulted in a higher percent error compared to displacement. I expected calculation method to be more accurate but I do believe in the possibility of me measuring incorrectly.

2. Indicate the most appropriate type of glassware to use when measuring out the following: (2 points each blank) Note: This section is **not** related to the data <u>you</u> generated in this lab.

	Type(s) of glassware that should be used
Exactly 8.00 mL of solution	, Pipette
Exactly 15.0 mL of solution	Pipette
Approximately 50 mL of DI water	Graduated Cylinder , Buret

a. What is the more precise glassware to measure liquids, a beaker or graduated cylinder? Graduated Cylinder

3. Consider the measurements in Table 5 below, recorded for a *rectangular* metal sample using ***an analytical balance*** and ***a ruler*** similar to those used in this experiment. (2 points each) Note: This section is **not** related to the data <u>you</u> generated in this lab.

Table 5. Measurements of a Rectangular Metal Sample

	Experimental Measurements
Mass of Metal Sample	95.524 g
Length of Metal Sample	4.60 cm
Width of Metal Sample	1.40 cm
Height of Metal Sample	1.30 cm
Volume of Metal Sample	8.37 cm
Density of Metal Sample	11.4 g/cm

a. Calculate the volume of the rectangular metal sample from Table 5 above. Record the volume on the blank in Table 5 above. Show your calculation.

$4.60\,\text{cm}\,(1.40\,\text{cm})(1.30\,\text{cm}) = 8.37\,\text{cm}$

b. Calculate the density of the rectangular metal sample from Table 5 above. Record the density on the blank in Table 5 above. Show your calculation.

$\frac{95.524\,\text{g}}{8.37\,\text{cm}} = 11.4\,\text{g/cm}$

c. Indicate two possible experimental errors with the measurements recorded in Table 5 above. (Complete sentences)

Reading the measurement incorrectly and Equipment can be Calibrated incorrectly leading to measurement error.

d. In coherent, complete sentences, **explain and prove by calculation,** if and how the experimentally determined density in Table 5 would change if a centigram balance (±0.01 g) was used instead of a milligram balance (±0.001 g).

Hint: To answer this question, please note the difference in *precision* between the two analytical balances shown below. Round the mass to the correct number of decimal places, and calculate the resulting density with the less accurate mass before answering this question.
(8 points)

centigram balance:



Judging off of calculation when using either a centigram or a milligram balance the values are approximately the same depending on the amount of significant figures,

milligram balance:



$D = \frac{95.529}{8.37\ cm^3}$

$= 11.41218637\ g/cm^3$

$D = \frac{95.524\ g}{8.37\ cm^3}$

$= 11.41266427\ g/cm^3$